

SPHINX V2 – PROGRAMMABLE CORE POWER

Core: 0.75 V nominal, 5 W active, 1 GHz target. Range: 0.30–1.20 V. I/O: 1.8 V. Schematic for PCB layout.

Protected input and auxiliary supply

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PMBus core regulator, 0.30–1.20 V

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Fixed 1.8 V I/O and calibrated telemetry

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USB control and remote reset

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10 | Independent temperature and IO overvoltage protection

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Independent board control power. Pico is command-only; held supply and reset state survives Pico loss. Functionality: one kernel execution. Power: back-to-back loop, then read steady-state rail V/I/P.

#FLG1
PWR_FLAG

GND

#FLG2
PWR_FLAG

VIN_RAW

#FLG3
PWR_FLAG

VIN_FUSED

#FLG5
PWR_FLAG

AUX5V

#FLG6
PWR_FLAG

PICO_VSYS

#FLG7
PWR_FLAG

0V75

#FLG8
PWR_FLAG

1V8

#FLG9
PWR_FLAG

CORE_DRTN

#FLG10
PWR_FLAG

CORE_AVIN

12 V input, eFuse, PMBus core regulator, current shunt, socket VCORE; fixed 1.8 V I/O. Core/I/O telemetry; Pico USB control; 12+12 capacitor population; hardware power shutdown latch and STOP. Valid VIO: reset defaults released; explicit button/ZedBoard/Pico reset only. PG and VIN12_VALID are monitors.

10 A continuous design target. Characterize voltage, frequency, power, and temperature per DUT. Change feedback scale with outputs disabled. Layout requirements: docs/LAYOUT_HANDOFF.md.

Author: Yuchen Mei
12 V nominal input | VCORE 0.30–1.20 V | VIO 1.80 V
Review package: PCB, PI, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board

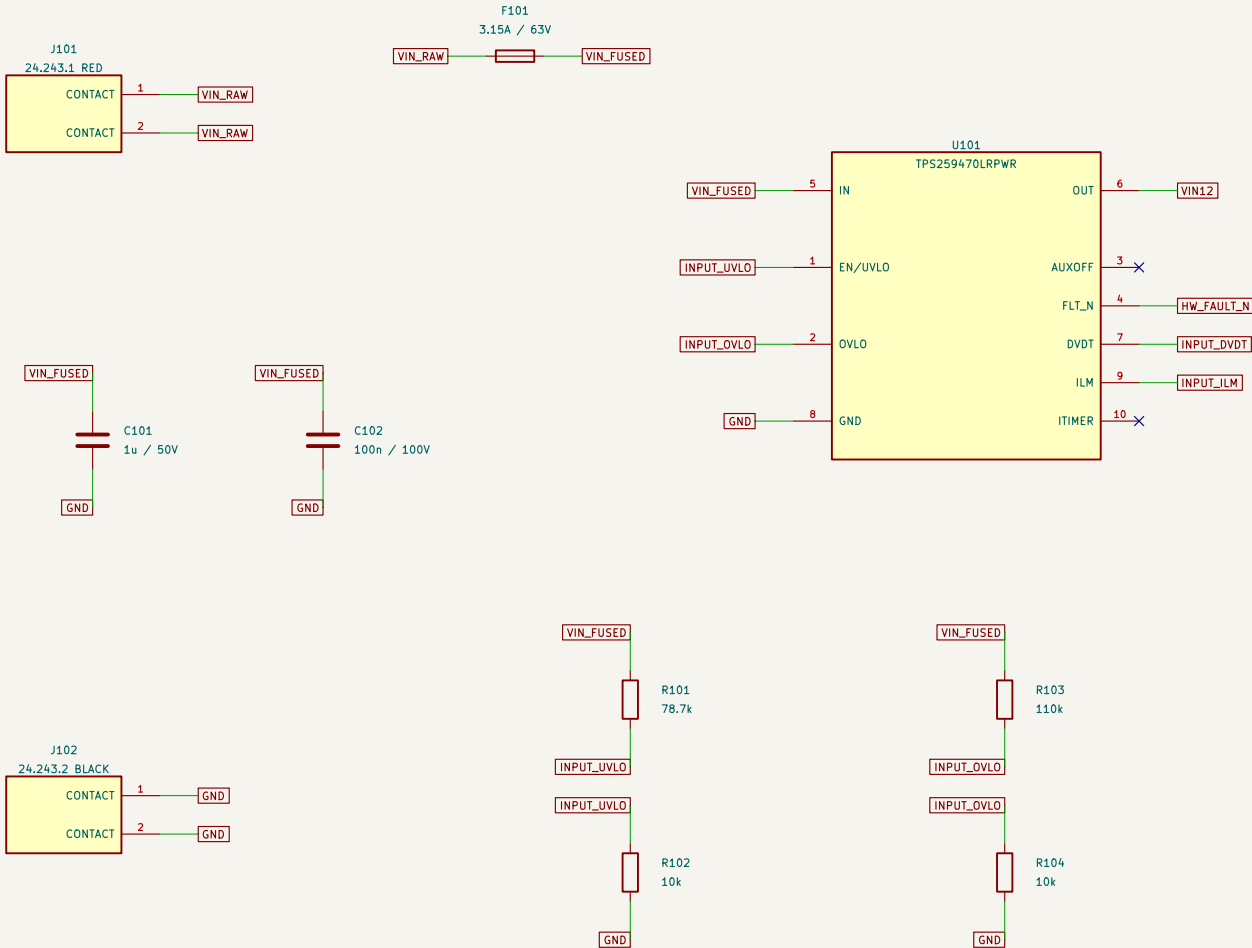
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File: sphinx_power_v2.kicad_sch

Title: Sphinx V2 | Programmable power test platform

Size: A3 | Date: 2026-09-22 | Rev: V2
KiCad E.D.A. 10.0.5 | Id: 1/11

INPUT: use 12 V / 3 A bench supply. OVLO disconnects a steady 20 V mis-setting.

F101 is backup protection. U101 UVLO/OVLO auto-recover; U509 VIN12_VALID is monitor-only.

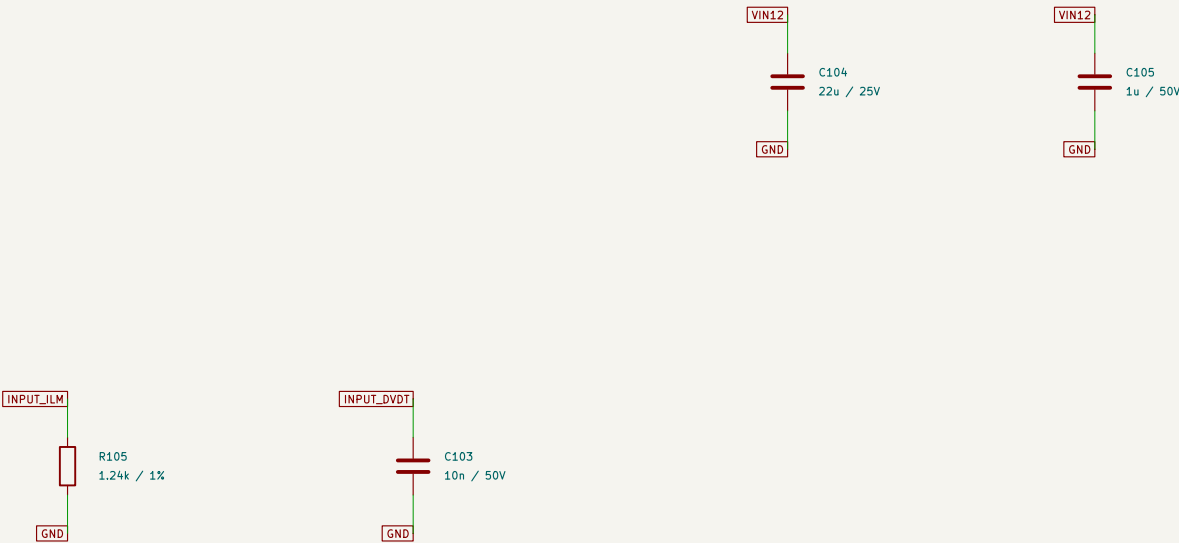


J101: +12 V supply positive.
J102: GND / supply negative.
Each socket: BOTH pads are common.
Use two full-current supply leads.

UVLO about 10.6 V rising; OVLO about 14.4 V rising. ILIM about 2.69 A (device tolerance applies).

Use 12V nominal; terminal target11.5–13V. Intrinsic UVLO interrupts power; eFuse FLT/control brownout can clear RUN_LATCH.

U101 OUT connects directly to VIN12.
Measure input current externally.
ILM connects only to local R105.



AUX5V = 0.596*(1+73.2k/10k) = 4.959 V nominal. U102 EN intentionally floating (internal enable).

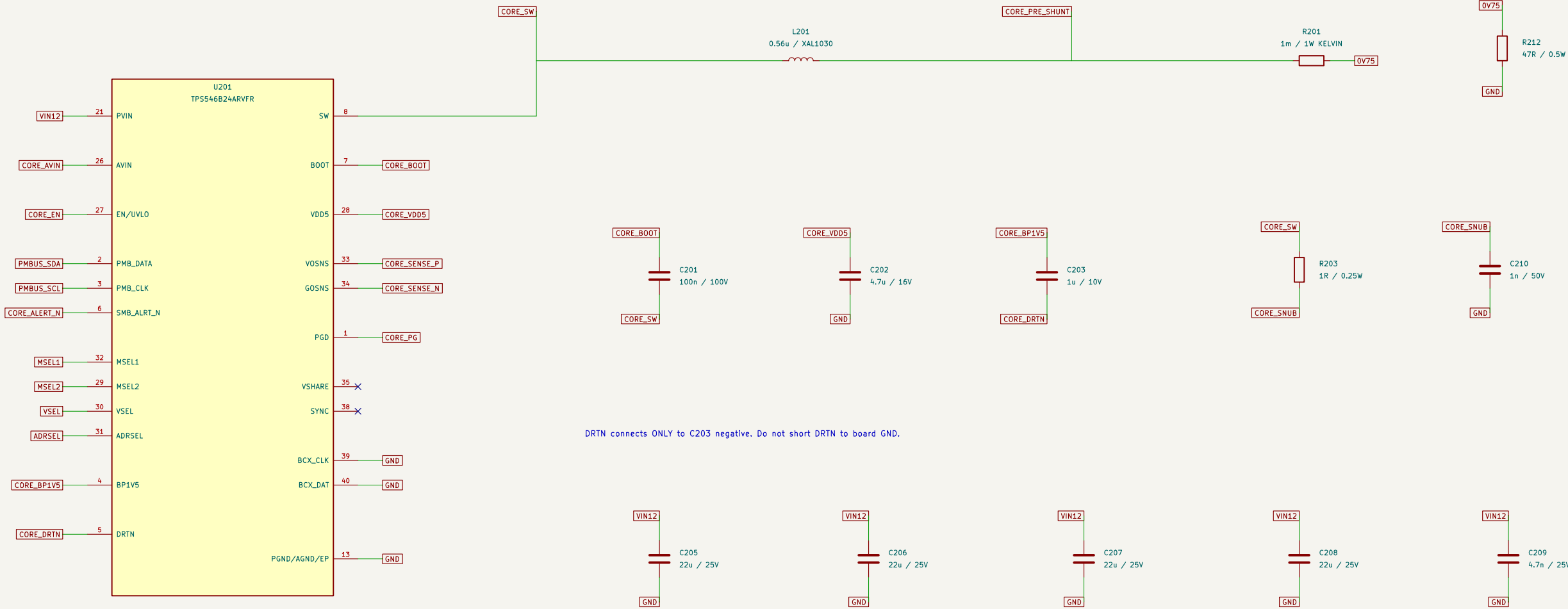
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12 V nominal input | VCORE 0.30–1.20 V | VIO 1.80 V
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Sheet: /Protected input and auxiliary supply/
File: 01_input_aux.kicad_sch

Title: 01 | Protected input and auxiliary supply

Size: A3	Date: 2026-09-20	Rev: V2
KiCad E.D.A. 10.0.5		Id: 2/11

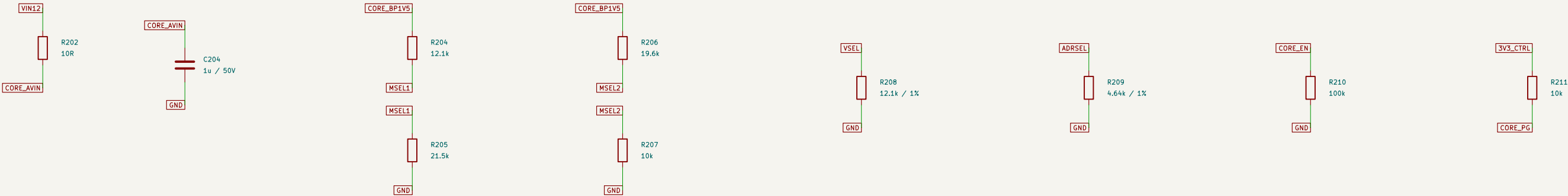
0.75 V nominal core; approximately 5 W during kernel execution. Board continuous-current design target: 10 A.

Boot 0.750 V, 650 kHz. Differential sense after R201 at socket. Output remains disabled until hardware run latch + retained board enable.



DRTN connects ONLY to C203 negative. Do not short DRTN to board GND.

Select 12+12 or 16+16 population; program and read back B1 with output OFF. Feedback scales: 1 and 0.5.



PMBus 7-bit address: 0x10, R209 = 4.64k to GND, no top resistor; SYNC auto-detect. Firmware must match.

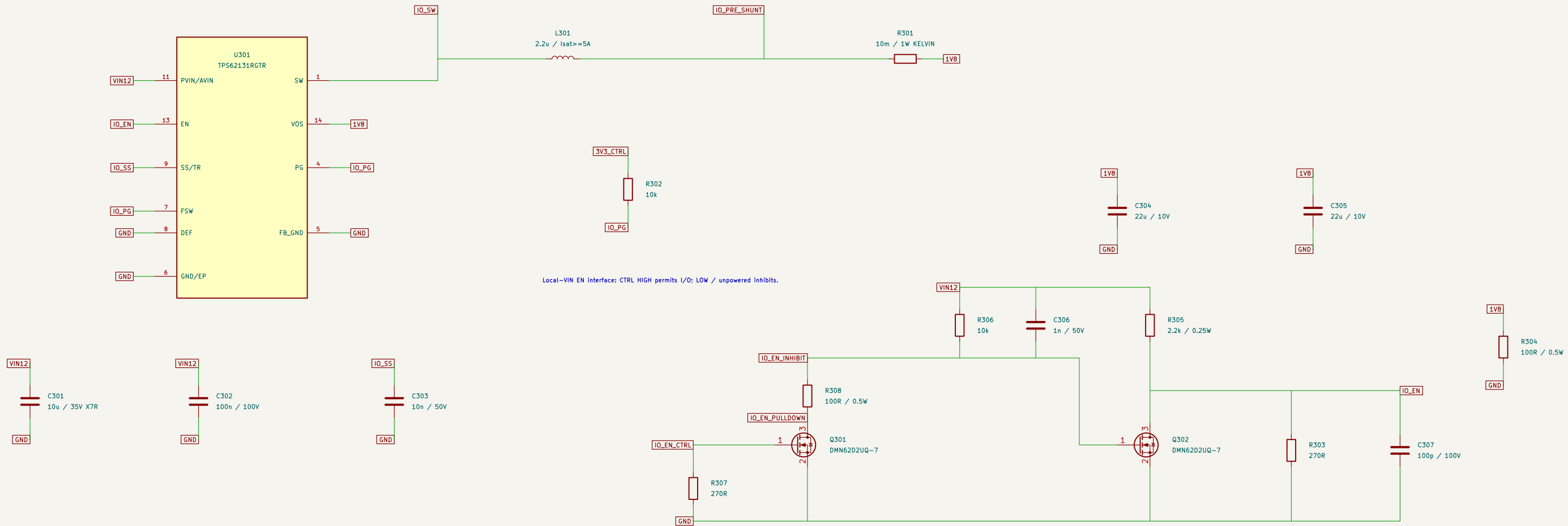
BCX_CLK/DAT and NC pin36 grounded for stand-alone operation, AGND37 and all PGND/EP pads share local return.

Author: Yuchen Mei
12 V nominal input | V_{CORE} 0.30–1.20 V | V_{IO} 1.80 V
Review package: PCB, Pl, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board
Sheet: /PMBus core regulator, 0.30–1.20 V/
File: 02_core_regulator.kicad_sch

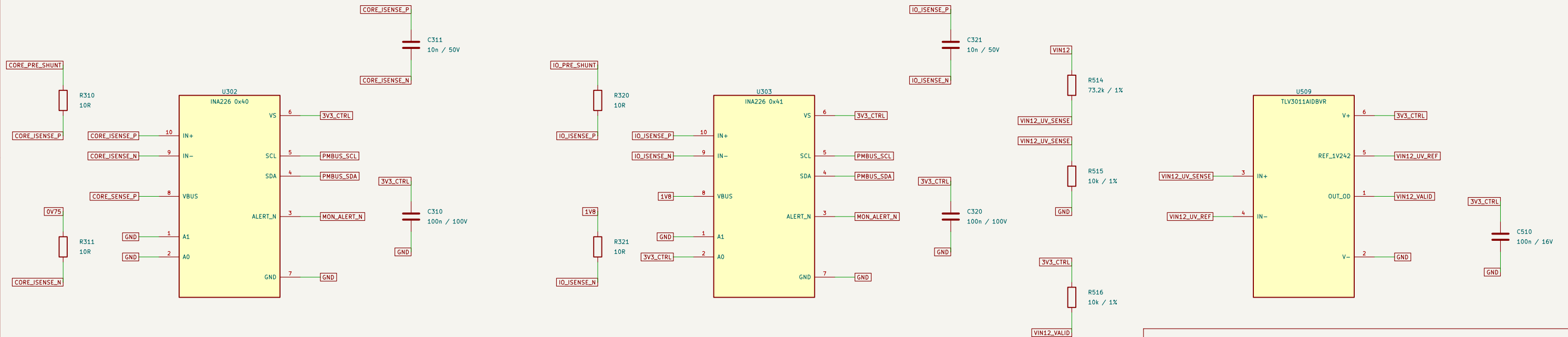
Title: 02 | PMBus core regulator, 0.30–1.20 V

Size: A3	Date: 2026-09-19	Rev: V2
KiCad E.D.A. 10.0.5		Id: 3/11

VIO: fixed 1.80 V / 3 A regulator capability; firmware default load budget 1 A. No I/O voltage sweep.



FWS tied to PG selects about 1.25MHz after startup. PG may float HIGH while OFF; U508 IO_VALID monitors VIO independently.



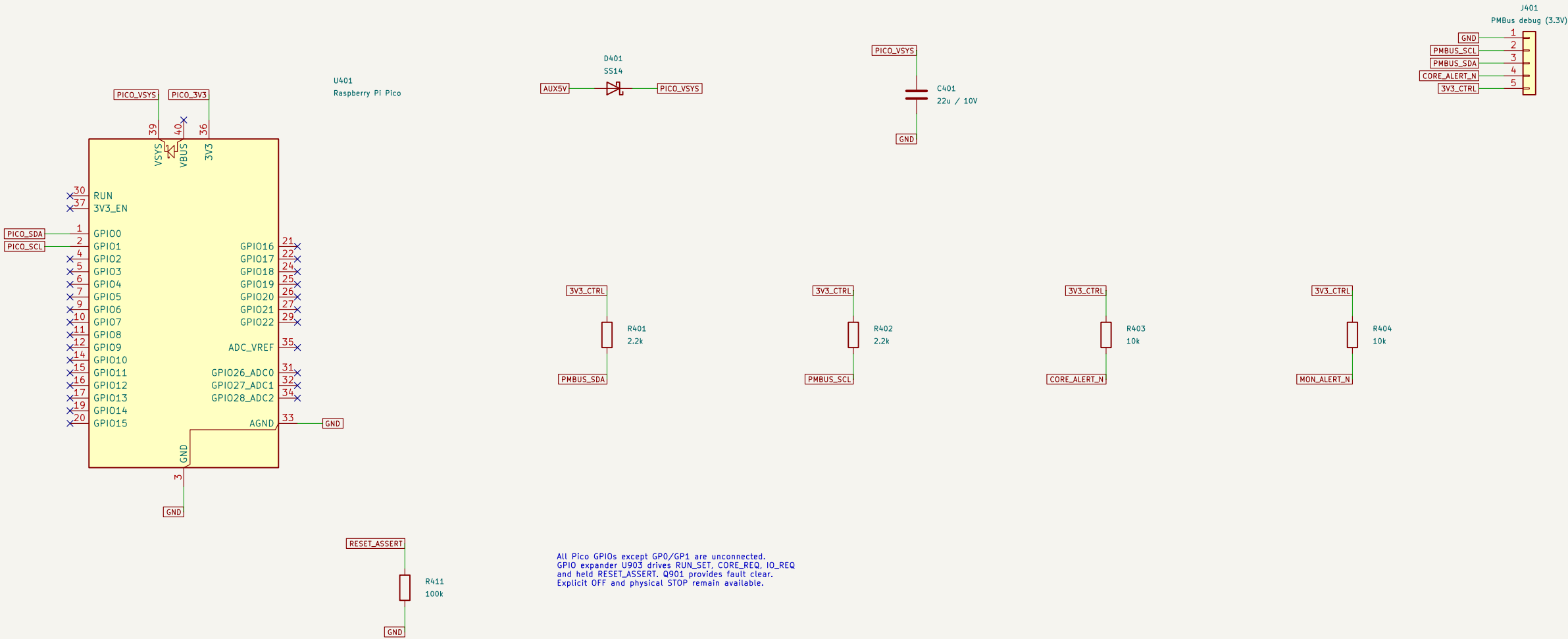
U509: VIN12_VALID to U903 P11, monitor-only; nominal threshold 10.33V. Intrinsic UVLO or control brownout can still interrupt rails.

INA226 averaging window 2.656ms: average rail telemetry only. No input-current ADC; bench supply reads total input.

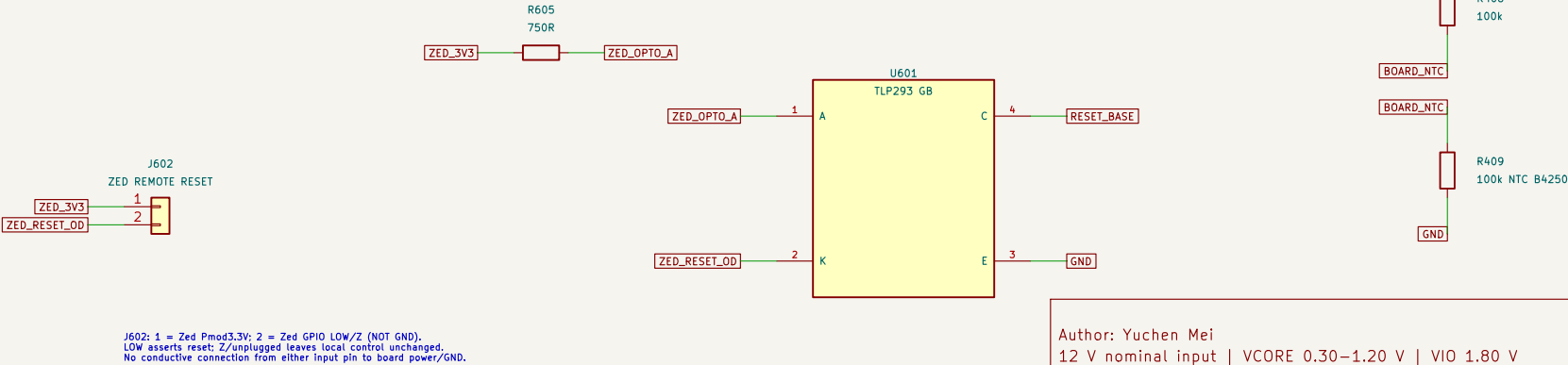
Author: Yuchen Mei
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Sheet: /Fixed 1.8 V I/O and calibrated telemetry/
File: 03_io_and_telemetry.kicad_sch
Title: 03 | Fixed 1.8 V I/O and calibrated telemetry
Size: A3 | Date: 2026–09–22 | Rev:
KiCad E.D.A. 10.0.5 | Id:

Raspberry Pi Pico module: USB CDC + MicroPython controller; no proprietary USB/PMBus adapter required.

USB commands: SET / ON / OFF / RESET / STATUS. Hardware controls: STOP and DUT RESET; reset is explicit only.



Pico is a command-only USB interface. Board state and protection are independent.
I2C is isolated by U902; U903 retains CORE / IO / RESET requests.
Pico boot, USB inactivity and communication errors do not change outputs.
See sheets09/10 and docs/AUTONOMOUS_POWER.md.



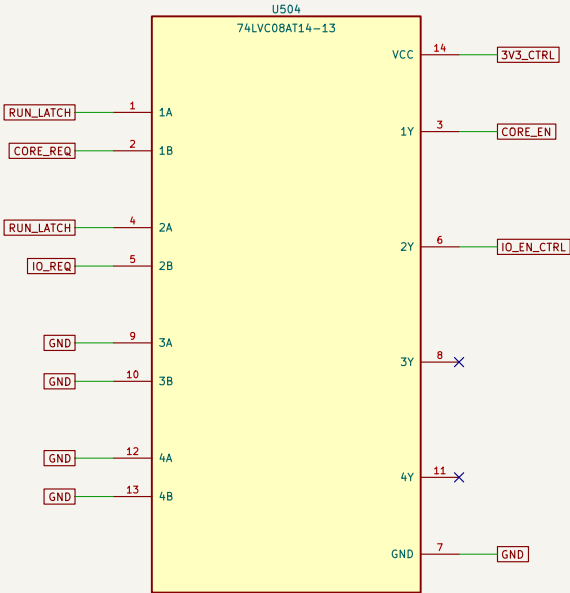
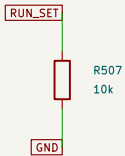
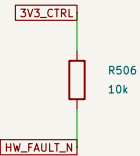
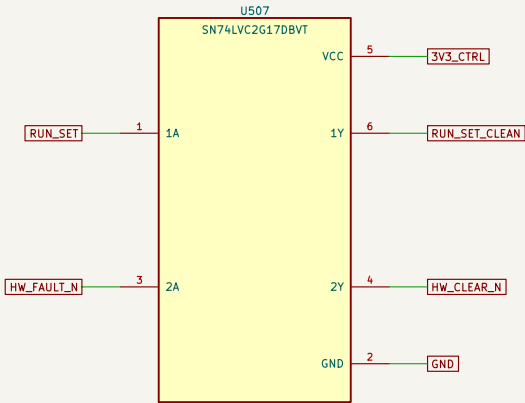
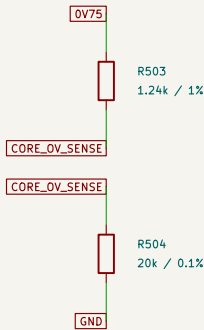
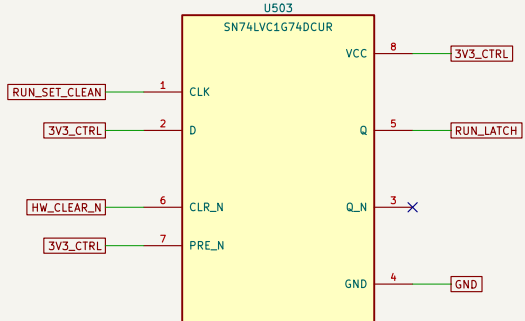
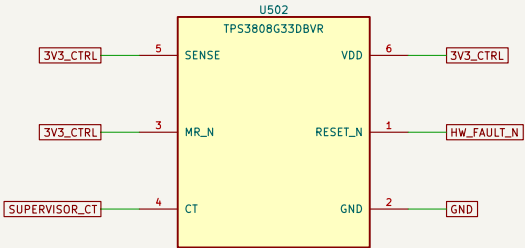
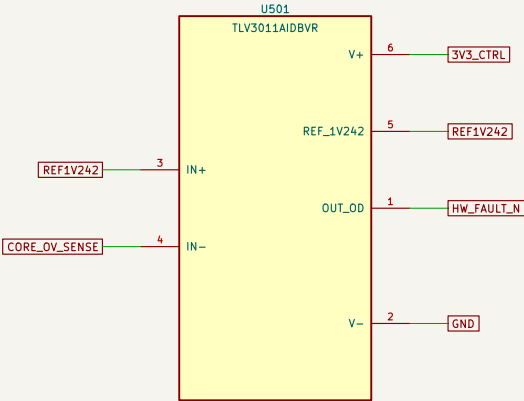
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Review package: PCB, PI, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board
Sheet: /USB control and remote reset/
File: 04_usb_control.kicad_sch

Title: 04 | USB control and remote reset

Size: A3	Date: 2026-09-22	Rev: V2
KiCad E.D.A. 10.0.5		Id: 5/11

ON sets the run latch. STOP, damage–risk faults or control brownout clear it; recovery requires a new ON command.

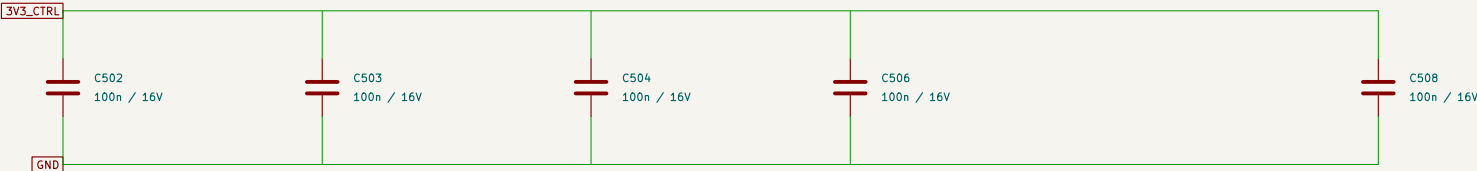
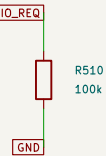
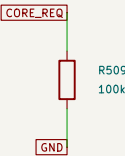
Core OVP and eFuse FLT retain hardware shutdown. VIN12_VALID, CORE_PG and IO_VALID are monitor–only.



U903 P03: one RUN_SET pulse inside ON, after configuration.
U903 P04 / Q901: LOW / high impedance only on HW_FAULT_N.
A stuck–high RUN_SET cannot restart a cleared latch.

DUT reset is independent of RUN_LATCH and PG.
At valid 1.8V, reset defaults LOW (released).
Only SW601, U601 or U903 RESET_ASSERT assert reset.

C502–U501: C503–U502: C504–U503.
C506–U504: C508–U507.
Place each bypass capacitor at its IC.



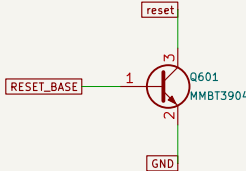
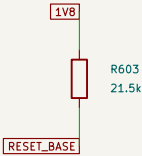
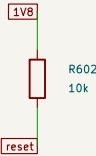
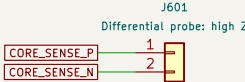
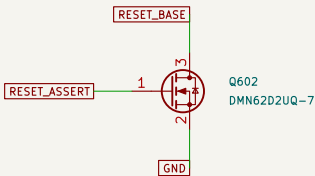
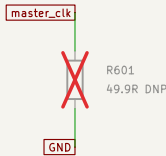
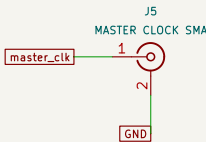
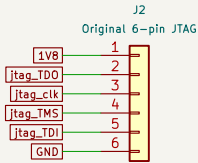
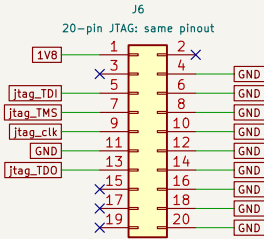
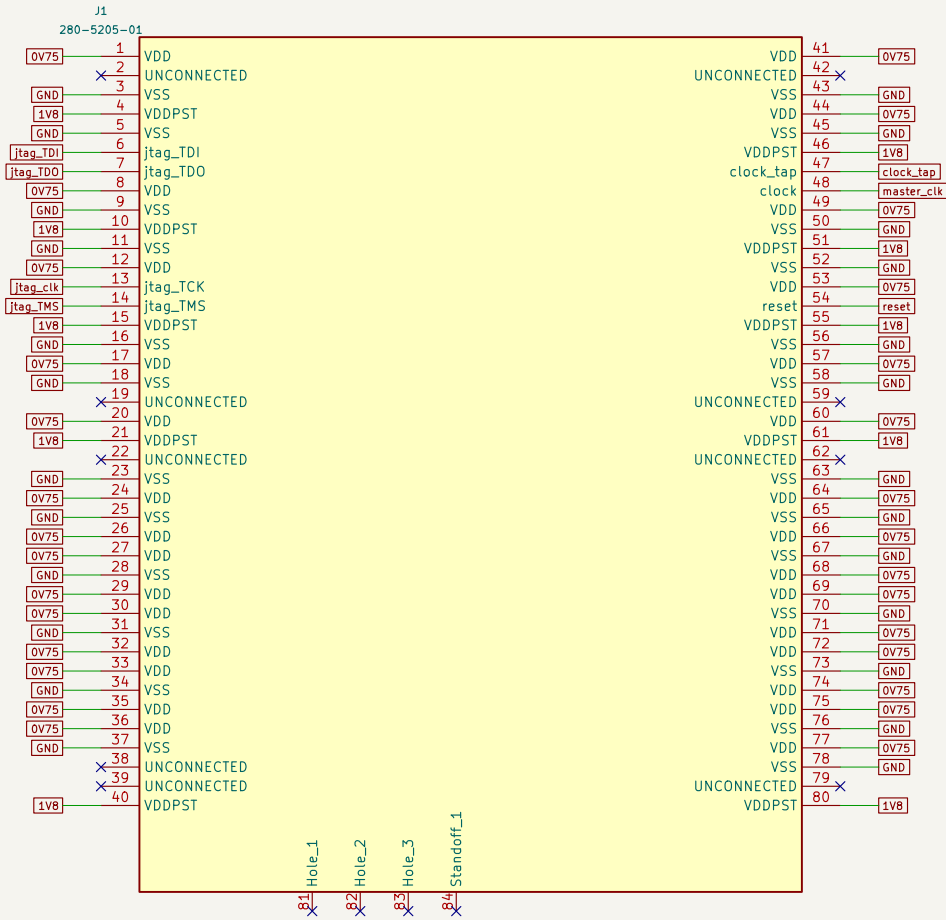
Author: Yuchen Mei
12 V nominal input | VCORE 0.30–1.20 V | VIO 1.80 V
Review package: PCB, PI, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board
Sheet: /Hardware power shutdown latch/
File: 05_hardware_safety.kicad_sch

Title: 05 | Hardware power shutdown latch

Size: A3 Date: 2026–09–22 Rev: V2
KiCad E.D.A. 10.0.5 Id: 6/11

J1: QFN80 socket, 29 VDD, 25 VSS, 10 VDDPST, 7 signals, and 9 NC positions.

Net 0V75 carries adjustable VCORE. NC positions and the center thermal probe have no electrical connection.



TP7 + TP607: by J1.54, <=2mm pad centres.
High-Z probe; no 50ohm termination.

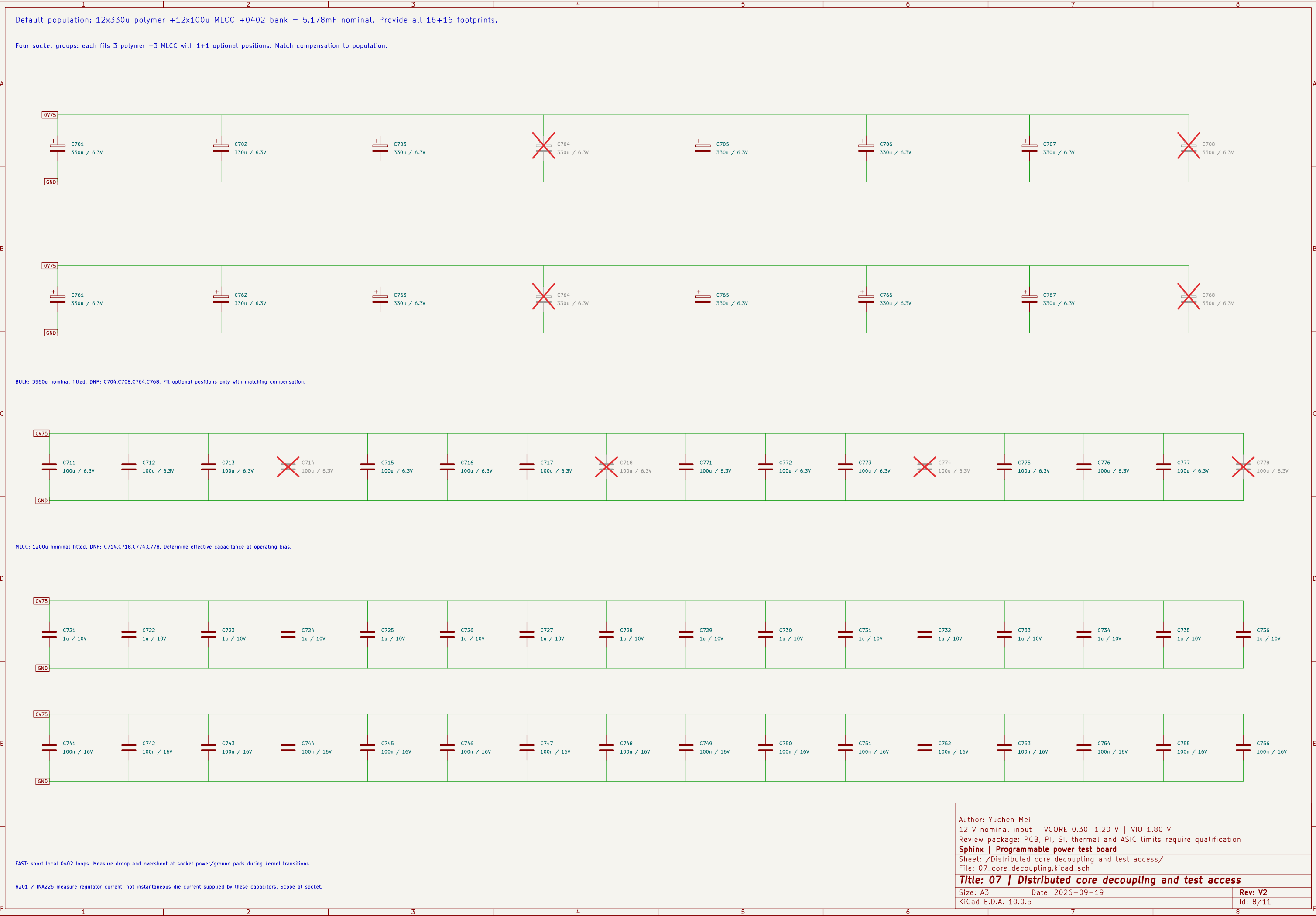
Valid VIO: reset LOW by default; any explicit request asserts HIGH.
U903 P02 LOW releases its held request. Place R411 at Q602 gate.

JTAG and clock source must use 1.8V logic and be high-Z while VIO is off. Do not connect a50ohm scope input across VCORE.

Author: Yuchen Mei
12 V nominal input | VCORE 0.30–1.20 V | VIO 1.80 V
Review package: PCB, PI, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board
Sheet: /Socket pinout and signal interfaces/
File: 06_socket_and_signals.kicad_sch

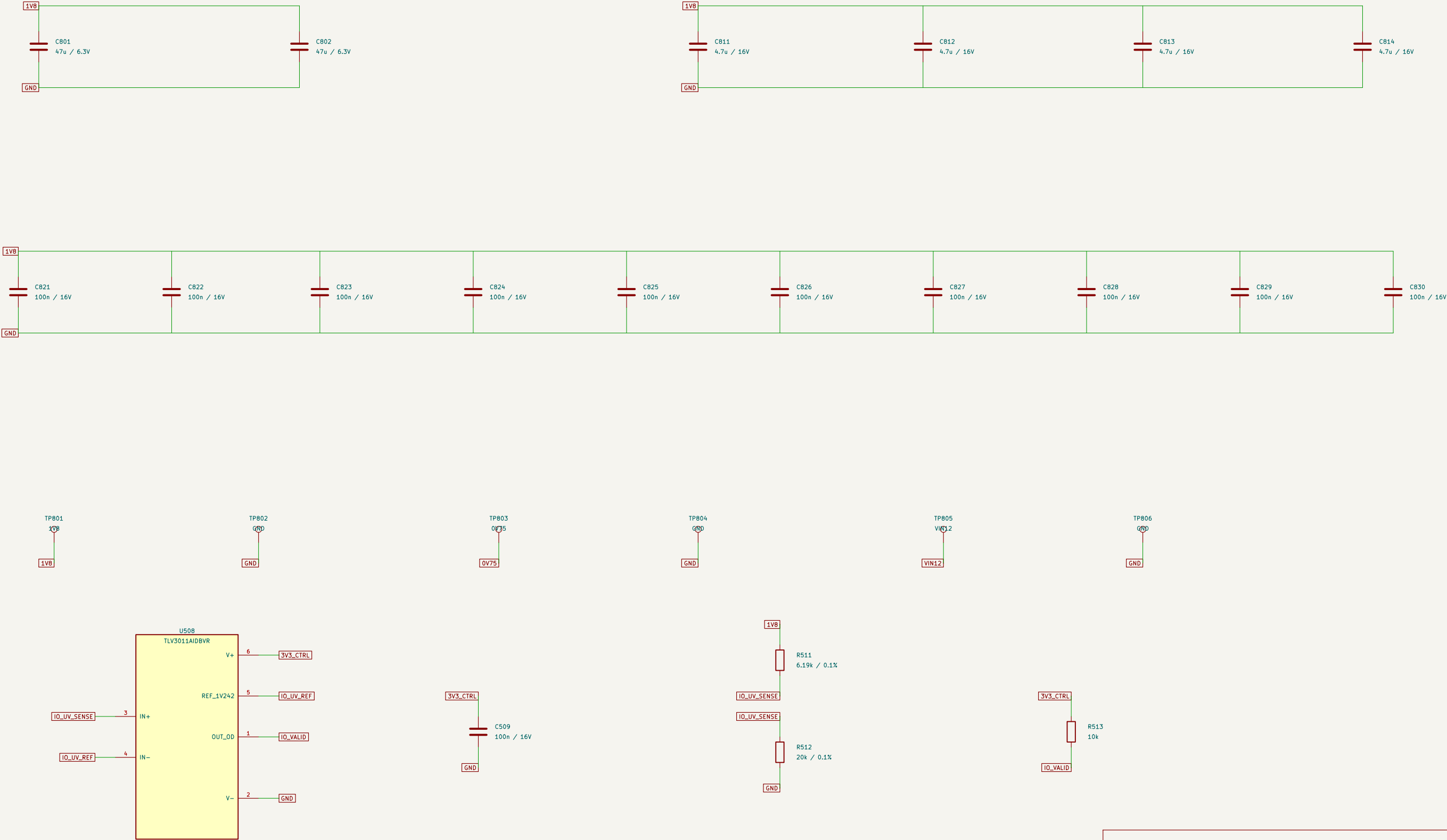
Title: 06 | Socket pinout and signal interfaces

Size: A3 | Date: 2026-09-22 | Rev: V2
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Fixed1.8V I/O supply: distributed local decoupling. Scope connections are high impedance.

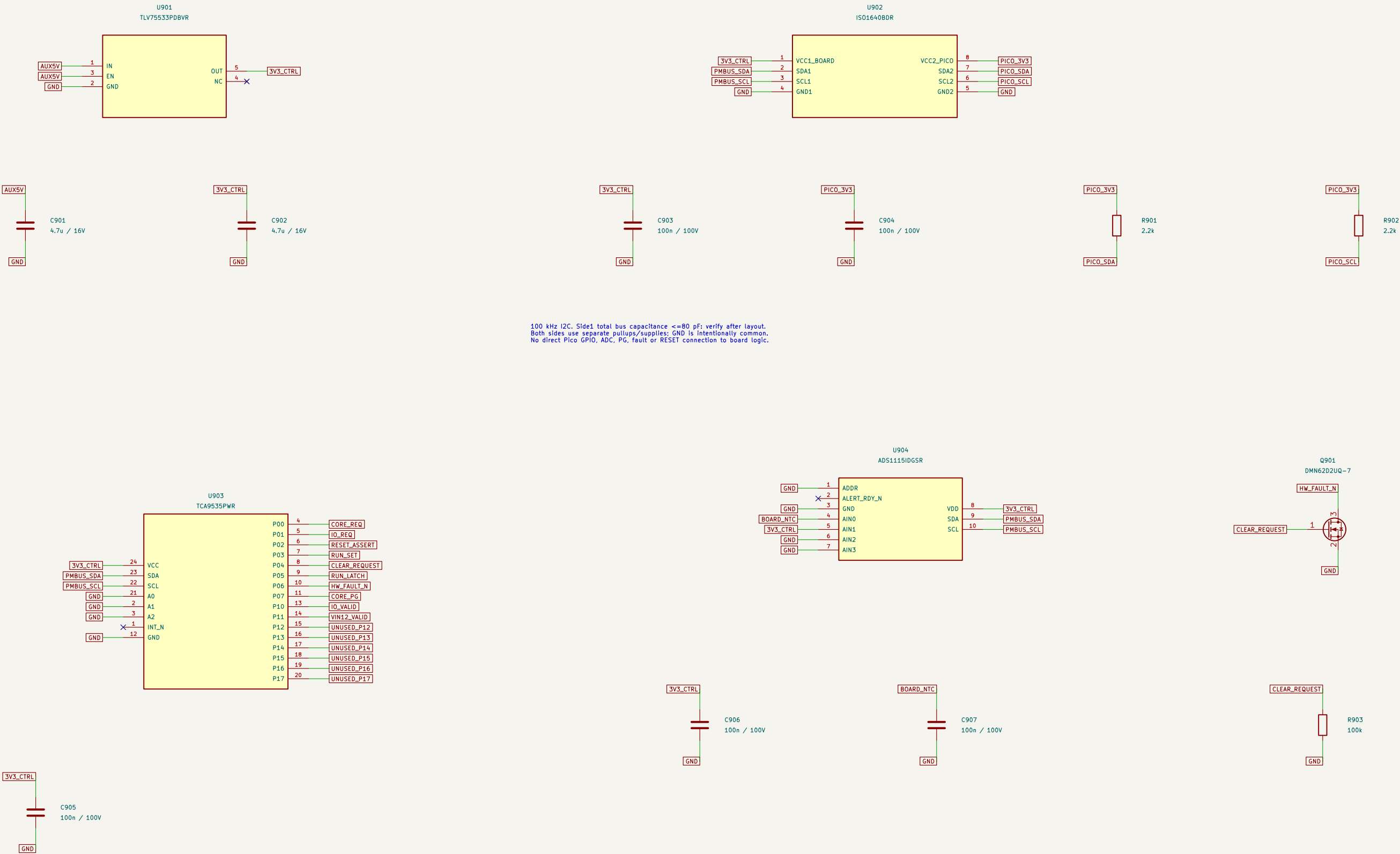
U508 measures VIO directly. IO_VALID reports to U903 P10 only; regulator IO_PG is only used by FSW.



IO_VALID threshold nominal1.626V; conservative static bounds about1.56–1.70V. Monitor-only; no reset or latching UV cutoff.

Place U508 divider at socket VIO; decouple each IC locally. Adjacent GND probe pads; high-Z probing; continuous signal return plane.

Board-powered state survives Pico reset, USB loss, firmware stop and Pico removal.



100 kHz I2C. Side1 total bus capacitance <=80 pF; verify after layout.
Both sides use separate pullups/supplies; GND is intentionally common.
No direct Pico GPIO, ADC, PG, fault or RESET connection to board logic.

NTC ratio uses AIN0 / AIN1 at +/-4.096 V PGA.
Board temperature is a local proxy; calibrate 100k divider loading.
75 C protection is Independent U1001, not ADC firmware.

Cold start only: write output latch=0 BEFORE direction=0xFFE0.
Pico boot/reconnect reads actual state; never initializes U903.
Completed commands leave RUN_SET=0 and CLEAR_REQUEST=0.
STOP / hardware faults clear U503 even with Pico absent.

Schematic revision; PCB and bench qualification required

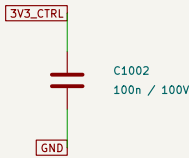
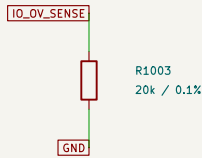
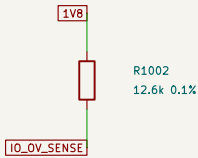
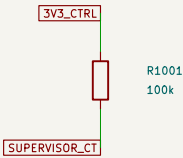
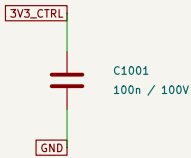
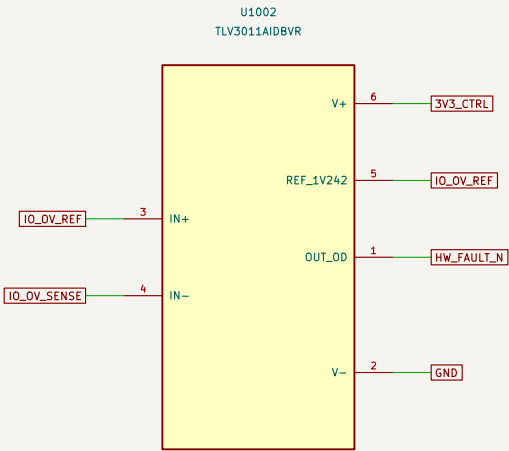
Sphinx | Programmable power test board

Sheet: /09 | Independent control supply and retained board state/
File: 09_autonomous_control.kicad_sch

Title: 09 | Independent control supply and retained board state

Size: A3	Date: 2026-09-22	Rev: V2
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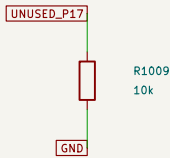
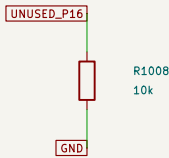
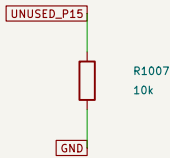
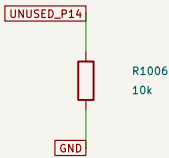
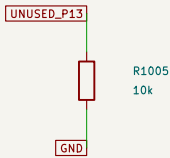
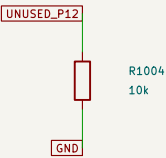
Hardware protection stays active with Pico unpowered or firmware stopped.



TMP302B: TRIPSET1=0,TRIPSET0=1 =>75 C; HYSTSET=0 =>5 C.
Place at the characterized board hotspot; it is not a die sensor.
Output startup35ms typical. U502 CT delay >=180ms; verify timing.
Qualify hot-start timing at supply and temperature corners.

IO OVP = $1.242 \cdot (1 + 12.6 / 20) = 2.02446$ V nominal.
Reference,offset,resistor and temperature errors apply.
Static screening: approximately1.943–2.106 V; measure actual trip.
Do not represent this as a precise2.025 V limit or ASIC safe voltage.

Unused expander input pull-downs; do not configure these ports as outputs.



Both open-drain outputs join HW_FAULT_N and asynchronously clear U503 through U507.
Fault release creates no RUN_SET edge; explicit ON is required for recovery.
Independent core OVP, control brownout, input eFuse FLI and physical STOP remain on sheet05.
Core regulator intrinsic OVP/OC/OT protects Core without Pico; a regulator-only fault need not clear IO.

Schematic revision; PCB and bench qualification required

Sphinx | Programmable power test board

Sheet: /10 | Independent temperature and IO overvoltage protection/
File: 10_independent_protection.kicad_sch

Title: 10 | Independent temperature and IO overvoltage protection

Size: A3 | Date: 2026-09-22

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